

Robot Logic and Conditional Coding

Piter Garcia

March 20, 2026

Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 5
- **Grade Hub:** Notes [../grade-5-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../school/Spring2026_TeachingSchedule_Ignite/grade-5/Spring2026_TeachingSchedule_Derek.md]
- **Workbook Sheet:** 5th in Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx
- **Lesson Focus:** Students move from basic robot control to conditional logic, using sensors, events, or logic choices to change what the robot does.
- **CS Alignment Strength:** Strong CS lesson
- **LaTeX Source:** robot-logic-and-conditional-coding.tex [./robot-logic-and-conditional-coding.tex]
- **Bib File:** robot-logic-and-conditional-coding.bib [./robot-logic-and-conditional-coding.bib]
- **Artifacts Folder:** artifacts/ [./artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 2 – 10:45-11:35 – Pickett, Day 3 – 10:45-11:35 – McGlashon, Day 5 – 10:45-11:35 – Pecora
- **Main Lesson Family:** Robot Logic and Conditional Coding
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- what a condition means in code
- how robot behavior changes when a condition is true or false
- how to compare a simple sequence with a logic-driven program
- how to debug conditional branches and trigger conditions

CS Standards Alignment

- **1B-AP-10:** Create programs that include sequences, events, loops, and conditionals. Why it fits here: The lesson explicitly targets conditional logic in robot programming.
- **1B-AP-11:** Decompose problems into smaller, manageable subproblems. Why it fits here: Students separate condition, action, and test case while planning the task.
- **1B-AP-15:** Test and debug a program or algorithm to ensure it runs as intended. Why it fits here: Students must test both branches or outcomes of the condition.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 5 robotics sequence in the workbook and schedule.
- Supports New York-facing problem solving and computational reasoning through branching logic and debugging.
- This lesson is a strong CS packet because students are no longer only sequencing movement; they are reasoning about logic and alternate outcomes.

Lesson Content and Concepts

- conditional
- branching
- sensor input
- test case
- debugging

Evidence / Proof of Instruction

- **Transcript-derived note:** 260227 Grade 4 5 Shared Conditional Sphero [../transcript-derived-lessons/260227-grade-4-5-shared-conditional-sphero.md] The shared Grade 4-5 note explicitly frames the work as conditional coding with Sphero robots.

- **Transcript-derived note:** 260313 Mixed Shared Sphero Minecraft [../transcript-derived-lessons/260313-mixed-shared-sphero-minecraft.md] Later mixed notes show students working across robot logic and Minecraft-style command thinking, reinforcing the logic focus.
- **Source transcript:** 260227 Grade 4 5 Shared Conditional Sphero [../transcripts/260227-grade-4-5-shared-conditional-sphero.md] The source transcript supports that conditions and robot responses were taught explicitly, not inferred later.
- **Likely student proof:** Robot runs that behave differently based on a sensor or condition, plus explained fixes, count as direct evidence.

Assessment Evidence

- Student can explain what condition the robot is checking.
- Student can build or modify a conditional block sequence.
- Student tests and explains at least one failure or false trigger.

UDL and Inclusion Supports

Multiple Representation

- Show condition language with if/when visuals and concrete trigger examples.
- Use a two-column chart: if this happens / then robot does this.
- Demonstrate one true-case and one false-case outcome.

Multiple Action / Expression

- Allow students to predict with cards before coding.
- Provide prewritten conditional sentence frames.
- Use paired testing roles so one student watches the condition while the other launches code.

Multiple Engagement

- Use visible cause-and-effect robot behavior to make logic concrete.
- Keep tasks framed as puzzles with more than one test round.
- Invite students to guess what the robot will do before it runs.

How We Are Already Making It Inclusive

- The transcript trail shows that students were already beyond basic movement coding and into more advanced reasoning.
- Robot tasks provide embodied feedback, which supports students who need visible evidence of logic.
- The lesson already appears to use partner structures and repeated testing cycles.

How to Improve Inclusion Next Time

- Add a branch organizer so students record both possible outcomes before launch.
- Create one simplified conditional task and one enrichment task in the same packet.
- Capture a short video of a working true/false branch in **artifacts/**.
- Add sentence stems for students explaining why the robot chose one action instead of another.

Materials and Setup

- Robots and charging setup
- Student devices
- Conditional challenge cards
- Sensor trigger objects or controlled environment

Teacher Moves / Script / Routines

- Model a non-conditional program first, then show what changes with a condition.
- Ask students to test both outcomes instead of only the successful one.
- Use debug talk that distinguishes wrong code from wrong trigger conditions.
- Close with a comparison between sequence-only and condition-based solutions.

Common Errors and Debugging

- Students may think a condition is just another step, not a decision point.
- Sensors can create inconsistent results if the environment is uncontrolled.
- Students may test only one branch and assume the program works.

Extensions and Simplifications

- Extension: chain more than one condition or compare two conditional solutions.
- Simplification: use one obvious trigger and one obvious response.
- Extension: ask students to explain the logic in plain language before showing the blocks.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References